

APPM Project 2

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1 Introduction

Markov Chains use probabilistic rules to predict the state of a system on a given day. These systems are memoryless, meaning the state of the next time step depends only on the current state. This system is useful in modeling the spread of disease. Using different states and pathways, we can determine the likelihood of being in each state on a given day. Determining how a vaccination would affect the spread of disease over time is also possible, which is discussed in more detail after the fundamentals are introduced.

2 Modeling Small Population Epidemics with Markov Chains

2.1 Graphical Representation of Markov Chain for SEIR Model

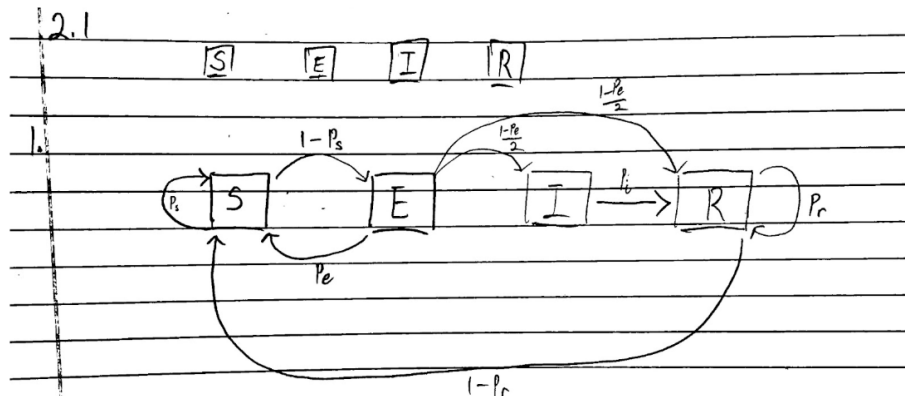


Figure 1: SEIR Model

2.2 Transition Matrix for SEIR Model

Given that $P_s = 0.7$, $P_e = 0.4$, $P_i = 1$, $P_r = 0.8$, by potting these values back to 2.1's graph, we will get that:

$$P = \begin{pmatrix} 0.7 & 0.4 & 0 & 0.2 \\ 0.3 & 0 & 0 & 0 \\ 0 & 0.3 & 0 & 0 \\ 0 & 0.3 & 1 & 0.8 \end{pmatrix}$$

2.3 Probabilities the Individual will be in Each State after One Iteration

When an individual is exposed to the disease

$$\vec{x}_0 = (0 \quad 1 \quad 0 \quad 0)$$

We take the transverse of the vector $\vec{x}_0$, which becomes

$$\vec{x}_0 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

Then we can compute the state after one day by multiplying it by P from Section 2.2
 $\vec{x}_1 = P * \vec{x}_0$

$$\vec{x}_1 = P * \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\vec{x}_1 = \begin{pmatrix} 0.4 \\ 0 \\ 0.3 \\ 0.3 \end{pmatrix}$$

Therefore, the probabilities are:

$$S = 40\%, I = 30\%, R = 30\%$$

2.4 Probability of Being in State R after 5 Days

First we start with the initial state vector

$$x_0 = \begin{pmatrix} 1.0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

and the transition matrix P

$$P = \begin{pmatrix} 0.7 & 0.4 & 0 & 0.2 \\ 0.3 & 0 & 0 & 0 \\ 0 & 0.3 & 0 & 0 \\ 0 & 0.3 & 1 & 0.8 \end{pmatrix}.$$

We want $P(R)$ after 5 iterations, and the equation will look something like:

$$x_5 = P^5 x_0.$$

By plotting it into Matlab we will get that

$$x_5 = 0.3408 = 34.08\%$$

Therefore, the probability of being in state R after 5 days is approximately 34.08%. (For code see Appendix A)

3 Stationary Probability Distributions

3.1 Probability of Being in Each State on a Given Day

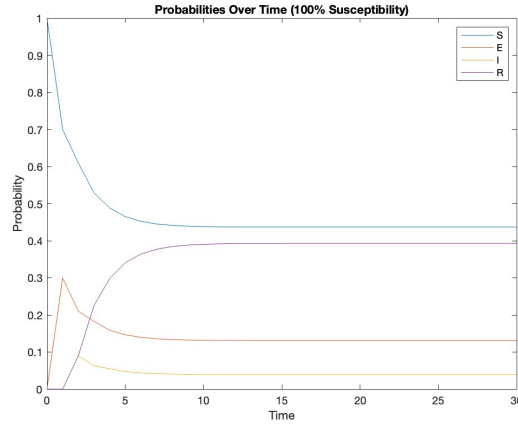


Figure 2: Probability of each state over time after 31 iterations of the SEIR Markov Chain

From the data provided in Figure 1, we now know the stationary distribution to within a reasonable error when each member of the population is initially susceptible. From the graph we can see that the probability for each state converges to different values over time, susceptible state converging to the highest value and infected converging to the lowest. 31 iterations is more than enough to observe this trend as the values stabilize after about 10 iterations. The stationary distribution over time when initial susceptibility is 1 is $\vec{x}_{31} = (0.4367 \ 0.1310 \ 0.0393 \ 0.3930)$. (For code see Appendix B)

3.2 Probability of Each State with 85% Chance of Exposure

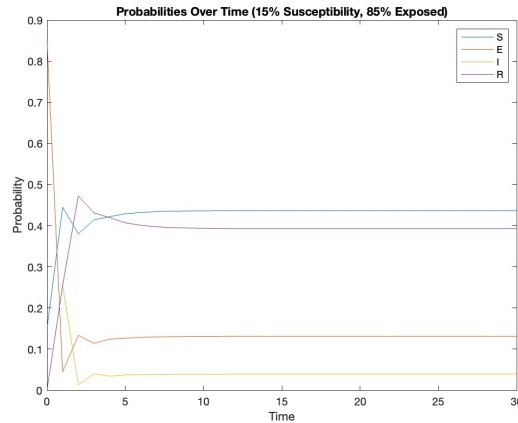


Figure 3: Probability of each state over time after 31 iterations with adjusted initial values

The stationary distribution of this SEIR chain remains the same even with adjusted initial values, reported similarly as $\vec{x}_{31} = (0.4367 \ 0.1310 \ 0.0393 \ 0.3930)$.
(For code see Appendix C)

3.3

When comparing Figure 1 and Figure 2, it is important to observe that the difference in initial state vectors had no effect on the stationary distribution reported for this SEIR chain. In fact, the initial state vectors only affected the first few steps. As opposed to Figure 1, a higher initial exposure percentage in Figure 2 resulted in stabilization of stationary values in much less time. Since the initial exposure

percentage is higher in Figure 2, the Markov Chain is able to spread over all states more quickly. 100% initial susceptibility creates a bottleneck, since it adds an extra step to get to the exposed state and therefore at least one extra step to spread out and reach stationary values.

3.4 Stationary Distribution of Markov Chain

3.4.1 Linear Combination of Eigenvectors of P

Since P is a transition matrix for Markov Chain, we assume that P has n linearly independent eigenvectors that forms basis for R^n . Let $v_1, v_2, v_3 \dots$ be eigenvectors $\lambda_1, \lambda_2, \lambda_3 \dots$

$$\vec{x}_0 = c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_n \vec{v}_n$$

Where $c_1, c_2, \dots$ are scalar coefficients.

3.4.2 "N-Step" State Vector of Markov Chain

Since

$$\vec{x}_N = P^N \vec{x}_0 = P^N (c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_n \vec{v}_n)$$

Therefore, by distributing P^N into each $c\vec{v}$, we get

$$\vec{x}_N = P^N c_1 \vec{v}_1 + P^N c_2 \vec{v}_2 + \dots + P^N c_n \vec{v}_n$$

Which can be written as

$$\vec{x}_N = \lambda_1 c_1 \vec{v}_1 + \lambda_2 c_2 \vec{v}_2 + \dots + \lambda_n c_n \vec{v}_n$$

3.4.3 Relationship Between Stationary Distribution and Transition Matrix Eigenvalues

Given that $|\lambda_1| \geq |\lambda_2| \geq \dots |\lambda_n|$ and $\lambda_1=1$, we know that

$$N \rightarrow \infty, \lambda_n^N \rightarrow 0$$

Except for $n=1$, because $\lambda_1=1$, and $1^N = 1$. Moreover, we'll get that

$$\vec{x}_\infty = \lim_{N \rightarrow \infty} \vec{x}_N = 1 c_1 \vec{v}_1$$

Which sums up to as N approaches ∞ , every number of eigen value that has $n > 1$ will disappear, leading to $c_1 \vec{v}_1$.

3.4.4 Stationary Distribution Graph

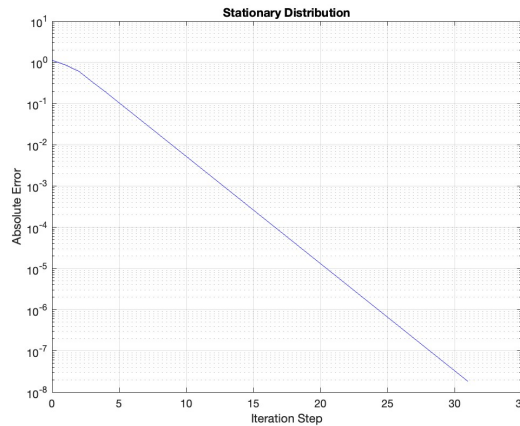


Figure 4: Stationary Distribution at each iteration step

From the error plot we can see that the slope appears to be a straight decreasing line, since it's a semilog graph, it means that the error is decreasing exponentially at each iteration step. Moreover, as iteration step increases, the error will become very small eventually, meaning that the state vector $\vec{x}_n$ is converged to the stationary distribution $\vec{x}_\infty$, which means that the Markov Chain is converging at an exponential rate.

(For code see Appendix D)

3.5 SEIR-Im Model

3.5.1 Graphical Representation of Markov Chain for SEIR-Im Model

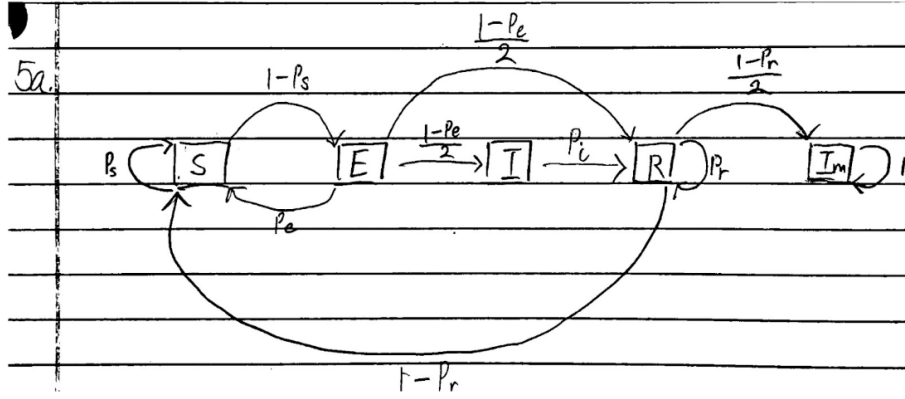


Figure 5: SEIR-Im Model

3.5.2 Transition Matrix for SEIR-Im Model

The transition matrix for the new model would look something like:

$$P = \begin{pmatrix} 0.7 & 0.4 & 0 & 0.1 & 0 \\ 0.3 & 0 & 0 & 0 & 0 \\ 0 & 0.3 & 0 & 0 & 0 \\ 0 & 0.3 & 1 & 0.8 & 0 \\ 0 & 0 & 0 & 0.1 & 1 \end{pmatrix}$$

3.5.3 Probability Over time

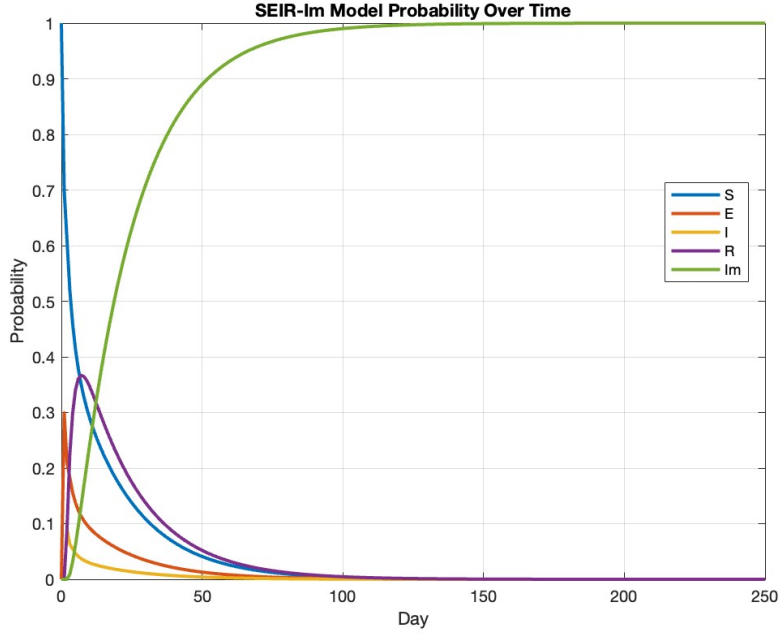


Figure 6: SEIR-Im Model Probability after 250 Day

We can see that the stationary in this SEIR-Im model is really different from SEIR model. Back in the SEIR model, the stationary distribution are: $S = 0.4367, E = 0.1310, I = 0.0393, R = 0.3930$. But in the SEIR-Im model, the stationary distribution are: $S = 0, E = 0, I = 0, R = 0, Im = 1$, showing that as time goes on, every state reaches zero except for Immune state, which means every state will reach immune state 100%, compared to SEIR model, there are still different percentage that one can be in and stay in that state specific state. To summarize, SEIR model gives out 4 different state that someone can be in, while SEIR-Im model gives out only 1 state that someone can be in as time goes on. (For code see Appendix E)

4 SEIR-VIm Markov Chain Analysis

4.1 Construct the Transition Matrix

The transition matrix P is:

$$P = \begin{bmatrix} 0.6 & 0.4 & 0 & 0.2 & 0 & 0 \\ 0.3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.3 & 0 & 0 & 0 & 0 \\ 0 & 0.3 & 1 & 0.8 & 0 & 0 \\ 0.1 & 0 & 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 0.5 & 1 \end{bmatrix}$$

4.2 Eigenvalue Multiplicity and Stationary Distribution

Multiplicity of the eigenvalue $\lambda = 1$ would still be 1, this shows that the stationary distribution is unique, meaning that every state has different chances to transit to other states, but this doesn't apply to Im state, if it lands in Im state, it's going to stay there. In other words, it means that Im state is the only state we can't leave, and other states leads to Im states eventually as time goes on

4.3 Probabilities of SEIR-VIm model after 31 days

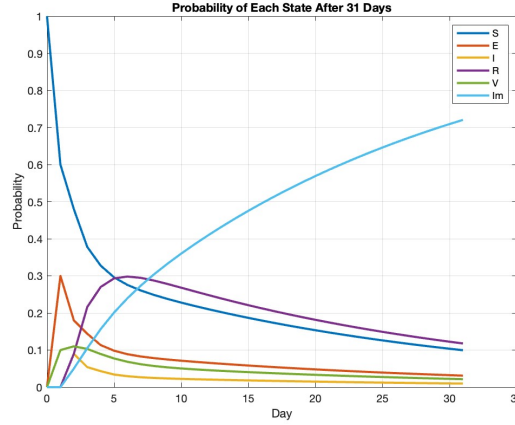


Figure 7: Evolution of state probabilities over 31 steps with initial state 100% susceptibility

The stationary distribution of this SEIR-VIm chain is again drastically different from its predecessors with the introduction of a "vaccinated" state. The stationary distribution is reported as $\vec{x}_{31} = (0.0996 \ 0.0311 \ 0.0097 \ 0.1178 \ 0.0216 \ 0.7203)$.

(For code see Appendix F)

4.4 Probability After 31 Days with Initial Condition of 33% Susceptibility and 67% Vaccinated

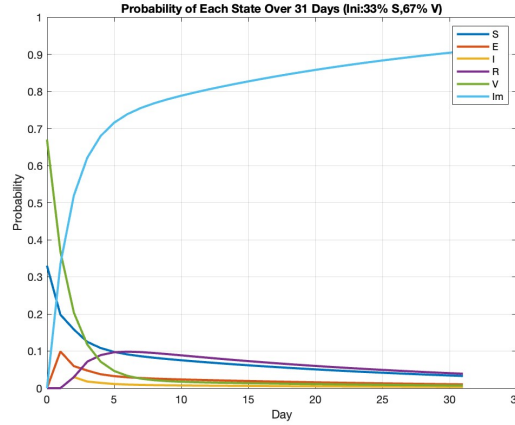


Figure 8: Evolution of probabilities over 31 steps with initial state: 33% susceptibility, 67% vaccinated

This time, the initial conditions do in fact influence the stationary values of the long-term convergence of the chain. This is due to "absorbing" states in the system, such as immunity and vaccination. The stationary distribution is reported as $\vec{x}_{31} = (0.0329 \ 0.0103 \ 0.0032 \ 0.0389 \ 0.0071 \ 0.9077)$.

(For code see Appendix G)

4.5 Dependence of Stationary Distribution on Initial State

The stationary distribution changes depending on the initial state vector. This behavior is because of the reducibility of the Markov Chain. Since the states S, E, I, R and V, Im do not communicate with each other, starting in one subset means the system will stay within subset. The transition matrix contains separate absorbing states (R and Im), and transitions between these states can't occur because of zero entries. This leads to multiple stationary distributions depending on the initial condition.

5 Conclusion

After exploring how Markov Chains can model the spread of disease through small populations, we determined that the chain converges to constant stationary values despite any modifications to the initial state of the system. However, the initial conditions will affect the rate of convergence: initial states closer to the "exposed" state will converge faster than those closer to the "susceptible" state. Moreover, when the "immune" state was introduced, it resulted in a much different state distribution, demonstrating how small tweaks to the structure of this model can result in major differences in the long-term outlook of our predictions. The Markov Chain is a powerful tool that effectively modeled disease dynamics using behavior in the steady-state as well as rates of convergence.